

Problem Set for 08 May 2025: A New Jordan form over $\mathbb{C}$

Important

Throughout, $\mathbb{C}$ is the base field. Define $|z| := \sqrt{z\bar{z}}$ for $z \in \mathbb{C}$.

Definition (Eigen-). Let $A \in \mathbb{C}^{n \times n}$. We say that (v, λ) is a eigenpair of A provided $A\bar{v} = \lambda v$ for some $\lambda \in \mathbb{C}$. The vector v is referred to as a eigenvector and λ a eigenvalue, correspondingly. For matrices, we say A is csimilar to B provided $A = PB\bar{P}^{-1}$ for some non-singular matrix P .

Note

You can firstly try Exercise 13-15.

Exercise 1 Characterise the eigenspace corresponding to the eigenvalue 0.

Warning

Eigenspaces associated with eigenvalues $\neq 0$ are not, in general, $\mathbb{C}$ -vector spaces.

Exercise 2 Demonstrate that if λ is a eigenvalue, then so too is $e^{i\theta} \cdot \lambda$. Furthermore, show that for any eigenvector v , there exists a eigenvector associated with a eigenvalue ≥ 0 .

Exercise 3 Suppose (v, σ) is a eigenpair of A ; then $(v, \sigma\bar{\sigma})$ is an eigenpair of $A\bar{A}$. In particular,

- if $A \in \mathbb{C}^{d \times d}$ possesses d eigenpairs which are linearly independent over $\mathbb{C}$, then $A\bar{A}$ is diagonalisable with all eigenvalues non-negative.

Now consider the following **partial converse statement**:

- A has no eigenvectors if and only if $A\bar{A}$ admits no eigenvalues in $\mathbb{R}_{\geq 0}$;
- $A \in \mathbb{C}^{d \times d}$ has d linearly independent eigenpairs over $\mathbb{C}$ if and only if $A\bar{A}$ is diagonalisable with non-negative eigenvalues.

Prove the **partial converse statement**: suppose that $(v, \sigma\bar{\sigma})$ is an eigenpair of $A\bar{A}$; then σ is a eigenvalue of A .

Tip

Construct an associated eigenvector of some $e^{i\theta}\sigma$, then invoke previous exercises.

Warning

When A has geometric multiplicity 1, one may simply take the eigenvector $e^{i\theta}v$; however, in general, $e^{i\theta}v$ is not a eigenvector. The construction closely resembles that found in **Exercise 6**.

Exercise 4 Elucidate **Exercise 3** (particularly the warning) with the following example:

$$A = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Exercise 5 This serves to complete **Exercise 2**. Suppose $\{(v_i, \lambda_i)\}_{i=1}^n$ are eigenpairs of A such that $|\lambda_i| \neq |\lambda_j|$; show that $\{v_i\}_{i=1}^n$ are linearly independent.

Exercise 6 We aim to identify the analogue of the Jordan canonical form in the c-version, namely, $X = PJP\bar{P}^{-1}$. We commence with a very special case:

- $P\bar{P} = I$ if and only if $P = Q\bar{Q}^{-1}$ for some invertible Q .

 Tip

Take $Q = a\bar{P} + bI$.

Exercise 7 The most elementary Jordan matrix is the diagonal matrix. Show that A is cdiagonalisable (i.e., $A = PD\bar{P}^{-1}$ for some P) if and only if $A\bar{A}$ is diagonalisable. In this scenario, $A\bar{A}$ has only non-negative eigenvalues.

 Note

In this diagonalisable case, one may also verify that $\text{rank}(A) = \text{rank}(A\bar{A})$; that is, the multiplicity of 0 is preserved.

Exercise 8 (Nilpotent part). Show that there exists a matrix P such that $P\bar{A}\bar{P}^{-1} = \begin{pmatrix} N & O \\ O & B \end{pmatrix}$, where N comprises nilpotent Jordan blocks and B is invertible.

 Tip

Consider the method we used last term to find $J_{s_i}(0)$'s. Useful hints can be found in **Exercise 11**.

Exercise 9 To analyse the Jordan form of A , we begin by studying the Jordan form of $A\bar{A}$. Demonstrate the following:

1. $A\bar{A}$ has a characteristic polynomial with real coefficients;
2. non-real Jordan blocks occur in conjugate pairs;
3. if $\dim \ker(\lambda I - A\bar{A}) = 1$ and $\lambda \in \mathbb{R}$, then $\lambda \geq 0$;
4. **(optional)** Generalise item 3 by showing that for any eigenvalue $\lambda < 0$, the dimension of the null space $\dim \ker(\lambda I - A\bar{A})^p$ is always even ($\forall p \geq 1$), implying that negative Jordan blocks occur in pairs.

 Warning

This is a straightforward exercise. Nonetheless, note that $A\bar{A}$ may possess negative eigenvalues.

Exercise 10 Show that there exists a matrix P such that $P\bar{A}\bar{P}^{-1}$ is block diagonal with the following summands:

1. (Jordan type): $J_s(\lambda)$ where $\lambda \geq 0$;
2. (Half-Jordan type): Let $H_{2s}(\lambda) := \begin{pmatrix} O & J_s(\lambda) \\ I & O \end{pmatrix}$.
 1. (For non-real eigenvalues): Since each complex Jordan block in $A\bar{A}$ appears in a pair, i.e., $\begin{pmatrix} J_s(z) & O \\ O & J_s(\bar{z}) \end{pmatrix}$, define $H_{2s}(z)$ as a block of $P\bar{A}\bar{P}^{-1}$ satisfying

$$H_{2s}(z) = \begin{pmatrix} O & I \\ J_s(z) & O \end{pmatrix}.$$

2. (For negative eigenvalues): For a negative eigenvalue λ of A , the Jordan blocks also appear in pairs (see **Exercise 9**).

We refer to this as the cJordan form of A . It is uniquely determined by A .

 Tip

The uniqueness (as we equal $H(z)$ and $H(\bar{z})$) of non-zero J - and H -blocks is governed by $A\bar{A}$; the uniqueness of the $J(0)$ -portion is determined by the rank sequence $\text{rank}(A)$, $\text{rank}(A\bar{A})$, $\text{rank}(A\bar{A}A)$, and so forth.

Exercise 11 (optional) (The uniqueness between the cJordan form of A and the Jordan form of $A\bar{A}$). The cJordan form is essentially unique.

- Find distinct matrices $A \neq B$ such that $A\bar{A} = B\bar{B}$;
- Suppose $A\bar{A} = B\bar{B}$ are invertible; then there exists a matrix P such that $P\bar{A}\bar{P}^{-1} = B$. In this case, we say A is scimilar to B .

In summary, A is csimilar to B if and only if the following hold:

1. $A\bar{A}$ is similar to $B\bar{B}$;
2. $\text{rank}(A\bar{A}A \cdots A(\bar{A})) \equiv \text{rank}(B\bar{B}B \cdots B(\bar{B}))$; that is, A and B possess the same $J(0)$ -portion in their cJordan forms.

Exercise 12 (optional) Utilise the cJordan form to show that any complex matrix A is csimilar to:

1. λA for $|\lambda| = 1$;
2. $\bar{A}$;
3. A^T ;
4. a real matrix;
5. a Hermitian matrix.

Exercise 13 (optional) Show that for any complex square matrix A ,

$$\det(AA^H + I) \geq \det(A\bar{A} + I) > 0.$$

 Tip

Apply the identity $AA^T + BB^T = RR^T$, and the Cauchy–Binet formula to establish the first inequality.

Exercise 14 (optional) Using cJordan forms, show that any complex matrix is the product of a symmetric matrix and a Hermitian matrix.

 Note

It is known that any real or complex matrix can be expressed as the product of two symmetric matrices (see exercises from the past semesters).

Exercise 15 (optional) Show that $\begin{pmatrix} O & A \\ \bar{A} & O \end{pmatrix}$ is similar to $\begin{pmatrix} O & B \\ \bar{B} & O \end{pmatrix}$, if and only if $\begin{pmatrix} O & A \\ -\bar{A} & O \end{pmatrix}$ is similar to $\begin{pmatrix} O & B \\ -\bar{B} & O \end{pmatrix}$.

 Tip

Equivalently, if and only if A and B are csimilar.